Fiscal Year:	FY 2013	**Task Last Updated:**	FY 03/01/2013
PI Name:	Bouxsein, Mary Ph.D.		
Project Title:	Maintenance of Skeletal Integrity in Spaceflight: Interaction among Partial Weight Bearing, Microgravity and Novel Therapeutics		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	02215-5400	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	04/01/2010	**End Date:**	01/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-135 NOTE: Received extension to 1/31/2014 per A. Chu/ARC (Ed., 8/19/13) NOTE: received extension to 9/1/2013 per NSSC (Ed., 4/23/2012)		
Key Personnel Changes/Previous PI:	There have not been any changes to key personnel		
COI Name (Institution):	Rutkove, Seward (Beth Israel Deaconess Medical Center)		
Grant/Contract No.:	NNX10AE39G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Despite intensive countermeasures, muscle and bone loss remain among the top medical challenges to extended space missions. Whereas much is known regarding the musculoskeletal responses to microgravity, little is known about the physiological effects of prolonged exposure to partial gravity environments, and whether musculoskeletal deterioration may be mitigated by exposure to partial gravity. To address this knowledge gap, we have developed a novel ground-based model that allows chronic exposure to reduced quadrupedal weightbearing in mice. Using this model will determine whether partial weight bearing attenuates bone and muscle atrophy relative to full hindlimb unloading, and will test the ability of novel spaceflight-compatible therapeutics, one anti-resorptive (antibody to RANKL) and one anabolic (antibody to sclerostin), to inhibit skeletal deterioration across the hypogravity spectrum. Outcome assessments will include in vivo bone density and body composition measurements, along with ex vivo measures of muscle weight, bone microarchitecture, and strength. Serum markers of bone turnover, histomorphometry and gene expression patterns will allow us to delineate the mechanisms underlying the observed skeletal changes. In sum, our proposed studies use both a standardized microgravity analog along with a novel partial gravity model to directly address a key element of the Bioastronautics Roadmap--Accelerated Bone Loss and Fracture Risk. Our ultimate objective is to mitigate the risk for space-induced bone loss and bone fracture associated with missions to low earth orbit, the moon, and eventually Mars. Successful completion of these studies will provide strong rationale for testing these pharmacologic countermeasures in human bed-rest studies and then leading to a trial in astronauts traveling to the International Space Station (ISS). Our findings will also provide insight into the prevention and treatment of skeletal fragility due to immobilization, stroke, cerebral palsy, muscular dystrophy, and spinal cord injury.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The current research has profound impact on Earth-based diseases and disorders resulting from insufficient skeletal loading, such as muscular dystrophy, cerebral palsy, stroke, and other neurodegenerative disorders. Our goal is to characterize the amount of skeletal loading necessary to maintain bone and muscle mass, and to determine the efficacy of novel therapeutics in preventing bone and muscle loss. If successful, this would provide novel insight for the rehabilitation of individuals exposed to musculoskeletal unloading or disuse. Our findings to date indicate that treatment with sclerostin antibody would be likely be effective in preventing disuse induced bone loss in a variety of clinical conditions, such as those listed above.
Task Progress:	To date, we have 1) determined the musculoskeletal effects of partial weight bearing in untreated animals; and 2) completed all testing of the ability of the anti-sclerostin antibody (Scl-AbII) to prevent bone loss across a spectrum of reduced loading. In addition, in a supplementary grant, we have studied the ability of sclerostin antibody to prevent skeletal deterioration during spaceflight by conducting an experiment on the last shuttle mission, STS-135 in July 2011. We were not been able to complete Aim 1 of our ground-based work as initially proposed due to our inability to obtain huRANKL mice needed to test the antibody to RANKL. Instead, we proposed to test the ability of a novel anabolic therapy, inhibition of ALK3 receptor signaling. In the past year we have completed the in vivo study to determine whether inhibition of ALK3 receptor signaling via a soluble mBMRPR1A –mfc fusion protein abrogates bone loss during disuse. In brief, we found that treatment with mBMRPR1A–mfc fusion protein not only inhibited the bone deterioration due to hindlimb unloading, but led to increased bone formation, as evidenced by gains in total body BMD and improvements in trabecular and cortical bone microarchitecture that surpassed or equalled those seen in the CON-VEH group. Further, dynamic histomorphometry of the trabecular bone compartment confirmed increases in bone formation rate, accompanied by reduced osteoclast number in the treated mice. In our supplemental grant, we proposed not only to study the ability of sclerostin antibody to inhibit bone deterioration in spaceflight, but also to conduct a companion hindlimb unloading study which matched the experimental protocol of the STS-135 flight (SF). This study was completed in the past year, mimicking the strain, sex, age, housing, food, and unloading duration of the spaceflight mice; mice were subjected to hindlimb unloading (n=15, HLU) and compared to concurrent normally-loaded controls (n=14,CON-HLU). Both SF and CON-SF lost a similar amount of weight, but this differed from the ground-based companion study where weight in CON-HLU did not change and HLU gained weight. Bone mineral density (BMD) of the forelimb was similarly reduced relative to CON by spaceflight and HLU. Hindlimb BMD increased in CON-SF (9.2%), whereas it was unchanged in SF (+1.3%). In comparison, hindlimb BMD declined in HLU (-3%) and increased in CON-HLU (+7.2%). Mass of all hindlimb muscles was lower in HLU compared to CON-HLU, with no differences in the biceps and triceps of the forelimb. In contrast, the only hindlimb muscles that were different between SF and CON-SF were the quadriceps and soleus, whereas biceps were significantly larger and triceps smaller in SF vs. CON-SF. Overall, these findings suggest that hindlimb unloading via tail-suspension leads to similar deficits in BMD as spaceflight; however, patterns of muscle atrophy were less consistent between HLU and spaceflight. Differences in forearm bone and muscle changes between HLU and spaceflight may reflect different use of the forelimbs during locomotion in spaceflight.
Bibliography Type:	Description: (Last Updated: 04/02/2026)
Abstracts for Journals and Proceedings	Bouxsein ML, Bateman TA, Hanson A, Pruitt T, Livingston E, Lemur M, Louis L, Ellman R, Spatz J, Warmington K, Tan HL, Hill D, Stolina M, Dwyer D, Ortega A, Maurya S , Lotinun, S, Baron R, Paszty C, Stodieck L, Ferguson V. "Sclerostin Antibody Treatment Improves Bone Mass, Microarchitecture and Mechanical Properties in Mice Exposed to Microgravity: Results from the STS-135 Shuttle Mission." 2012 Annual Meeting of the American Society for Bone and Mineral Research, Minneapolis, Minnesota, October 12-15, 2012. Oral Presentation. J Bone Miner Res 2012;27(Suppl 1):S20. Available at: http://www.asbmr.org/Meetings/AnnualMeeting/AbstractDetail.aspx?aid=7973771c-37d4-443c-8363-3e269b6d807e ; accessed 3/4/2013. , Oct-2012
Abstracts for Journals and Proceedings	Bouxsein ML, Bateman TA, Hanson A, Pruitt T, Livingston E, Lemur M, Louis L, Ellman R, Spatz J, Warmington K, Tan HL, Hill D, Stolina M, Dwyer D, Ortega A, Maurya S , Lotinun, S, Baron R, Paszty C, Stodieck L, Ferguson V. "Sclerostin Antibody Treatment Improves Bone Mass, Microarchitecture and Mechanical Properties in Mice Exposed to Microgravity: Results from the STS-135 Shuttle Mission." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. Oral Presentation. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013

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